

Hanlong Liu

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Master of Science in Computational Science & Engineering

Master of Science in Electrical and Computer Engineering

GPA: 4.0/4.0

Expected Graduation: 05/2027

The University of Michigan, Ann Arbor

Ann Arbor, MI

Bachelor of Science in Engineering (Computer Science Engineering)

08/2022 - 05/2025

Minor in Entrepreneurship (Ross School of Business)

GPA: 3.73/4.0

PATENTS

- Jain, D., **Liu, H.** *Earphone and Method of Use*. US Patent App. 63/706, 317 filed Oct 2024
- Developed a multi-modal earphone architecture integrating algorithmic calibration and real-time signal-processing pipelines.
- Hanlong Liu**, *A Kind of armature driver Five Unit Hi-Fi Earphone*, ZL 2019 2 0331833.4
- Designed a coordinated audio system using DSP-based tuning and multi-driver optimization algorithms to improve fidelity and stability.

INTERNSHIP

ByteDance – Lark (Enterprise Collaboration Platform)

Shenzhen, China

Intern, Artificial Intelligence Division

05/2023 – 08/2023

- Owned the LLM serving stack for a financial institution deployment; benchmarked vLLM vs HuggingFace TGI on A100 under 500-request load, achieving p99 latency 160ms vs 280ms, and selected vLLM for production.
- Identified KV cache block fragmentation as the primary memory bottleneck via nsys profiling; optimized block size (16 → 32 tokens), reducing peak VRAM by 18% without throughput regression.
- Evaluated tensor parallelism (TP=4) vs pipeline parallelism (PP=2) on 4×A100; selected TP for production workloads (<512 tokens) due to lower latency, while documenting PP advantages for long-sequence generation.
- Profiled FP16 vs BF16 on A100; BF16 showed 8% higher throughput on LLaMA 2 due to native hardware support, while FP16 caused attention overflow at sequence length >1024 — switched to BF16 for production.
- Integrated LLaMA 2 and ChatGPT API into Lark's internal platform; designed prompt templates and context window management for multi-turn enterprise dialogue, handling token budget constraints within 4096-token limits.

Development of a Patented Earphone Embedded Bone Conduction Technology

The University of Michigan, Ann Arbor

Research Intern, supervised by Prof. Dhruv Jain, Soundability Lab

03/2024- Present

- Designed and implemented a multi-stage software pipeline integrating DSP modules (filtering, calibration, cross-driver compensation), significantly improving audio quality and system reliability across diverse transducer types.
- Built a real-time audio processing engine featuring sound-source detection, adaptive noise suppression, and speech enhancement; optimized for low-latency, high-throughput streaming scenarios.
- Developed a 3-pass 3D reconstruction system using structured-light data (FaceID), incorporating point-cloud fusion, mesh alignment, and surface refinement to generate accurate anatomical models.
- Created a parametric CAD automation tool that converts reconstructed meshes into manufacturable geometries through rule-based feature generation, reducing manual modeling time by over 70%.
- Collaborated in a multi-disciplinary team and delivered end-to-end software modules in Python/C++ for audio processing, data pipelines, and 3D geometry processing.

Development of a AI Instructor Platform with RAG Knowledge Base

Georgia Institute of Technology

Graduate Research Assistant, supervised by Prof. Gregory Mihalik,

01/2026- Present

- Evaluated chunk-based RAG vs. Karpthy wiki synthesis over 14 lecture transcripts; selected wiki approach for superior cross-lecture retention at this scale; powered vector-search Q&A via in-space AI avatar
- Voice-cloned professor's delivery with Qwen TTS over ElevenLabs; tuned temperature and prosody until professor-validated; synchronized word-by-word across ~28 hours of narration
- Built Apple Vision Pro educational platform (RealityKit / Swift 6): 3D globe with startup case study pins, drag-momentum physics; validated with 5+ CREATE-X founders across US, India, and China
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TECHNICAL SKILLS

- Languages:** C++, Python, C
- ML / Frameworks:** PyTorch, CUDA, vLLM, HuggingFace Transformers, TensorRT, ONNX Runtime
- Systems:** Multi-GPU training/inference, Mixed Precision (FP16/BF16), Linux/Unix, Shell
- Developer Tools:** Git, Docker, NVIDIA profiling tools (nsys / nvprof)
- 3D & DSP:** Rhino 3D, AutoCAD, DSP tools, audio signal analysis
- Specialized:** Real-time inference optimization, 3D reconstruction pipeline, LLM fine-tuning, model serving & deployment